

Teaching Design with Consumer-ready Artificial Intelligence and Virtual Reality: A Case Study

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ABSTRACT

Design disciplines that require intensive ideation and demand spatial perception (e.g., architecture, interior, and industrial design) benefit from advanced computer graphics, interactive techniques, and human computer interactions. However, design students without a background in Computer Science (CS) face a steep learning curve and difficulties in experimenting with such advancements. Consumer-ready technologies, like Artificial Intelligence (AI) and Virtual Reality (VR), can leverage design skills such as risk-taking, iterating, and visualizing. Interior design students can translate their proficiency in concept writing and computer-aided design (CAD) software into AI prompt writing and VR digital modeling. This Engaging Education Technique and Assignment (EETA) features a nine-week lighting design project. Students use AI to ideate in 2 weeks, VR to visualize in 4 weeks, and 3D print to prototype in 3 weeks. Assessments for the 2022 class showed significant statistical evidence for creativity improvements with the use of consumer-ready AI and VR (but not with 3D printing).

KEYWORDS

Education, Design, Creativity, Pedagogy

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1 INTRODUCTION

The intersection between art (or arguably, its heuristic relative, design) and technology is no longer an “oxymoron” but a tautology [Chestney-Harvey, 2011]. Design disciplines with intensive ideation and demanding spatial perception, like architecture, interior, and industrial design, are benefiting from the advancements in computer graphics, interactive techniques, and human computer interactions [Yu et al., 2021, Van Der Maden et al., 2023]. However, design students without a background in CS face a steep learning curve and difficulties in experimenting with such technologies [Flechtner and Stankowski, 2023, Drampalou et al., 2023]. As an educator, the author encourages the use of consumer-ready technologies to augment interior design students’ ideation and spatial

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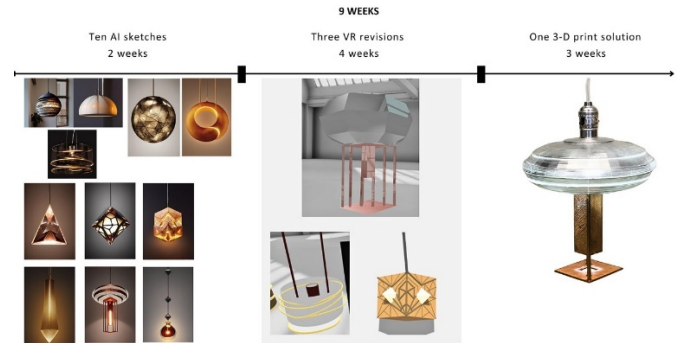


Figure 1: A synopsis of the nine-week project. Design by Gabrielle Wilson. Diagram by the author.

perception. Thus, this EETA submission provides perspectives on the other side of the screen, not as benefactors (creators) but as beneficiaries (users).

The author features a case study of integrating Midjourney¹, an AI text-to-image generator, and Gravity Sketch², a VR modeling platform, in a nine-week project for junior interior design students. Midjourney and Gravity Sketch have intuitive interfaces and procedures, with the former operating via a Discord chatbot and the latter running on Meta Quest 2 headsets. The project requires no computational knowledge (see Table 1). Instead, students leverage design skills such as risk-taking, iterating, and visualizing upon their proficiency in concept writing (translating to AI prompt writing) and CAD software (translating to VR digital modeling).

2 THE PROJECT

The author tailors Midjourney AI and Gravity Sketch VR to assist students in rapid ideation (2 weeks) and comprehensive spatial perception (4 weeks). Midjourney AI allows students to generate and experiment with a vast number of ideas before narrowing down to 10 initial sketches. Gravity Sketch VR then helped students transform two-dimensional (2D) sketches into three-dimensional (3D) models to understand better how their design will manifest in space with minimal costs of time and resources. Students further iterate the 3D models in terms of feasibility and aesthetic (e.g., alternating shapes and details). The project concludes with students 3D-printing and crafting materials for their design prototypes (3 weeks). Figure 1 summarizes this nine-week project. Students also submitted their outcomes to the annual Robert Bruce Thompson design competition.

¹<https://www.midjourney.com/home>

²<https://www.gravitysketch.com/>

Table 1: EETA metadata in tabular format

Summary	This nine-week project leverages risk-taking, iterating, and visualizing skills that are vital for design students in disciplines with intensive ideation and demanding spatial perception.
Learning Outcomes	Apply consumer-ready AI and VR to idea, iterate, and visualize a custom light fixture for the Robert Bruce Thompson annual design competition.
Classification(s)	Education; Human computer interaction (HCI)
Audience	This project is ideal for undergraduate students who has sufficient understanding of concept writing and CAD software but not CS knowledge (e.g., architecture, interior, and industrial design).
Dependencies	This project requires at least 5 dedicated VR headsets (e.g., Meta Quest 2) for a class of 20 students.
Prerequisites	Students need to complete lectures and exercises on lighting basics prior to this project.
Strengths	This project fosters risk-taking, iterating, and visualizing skills in students and prepare them for a technology-driven and ever-changing future.
Weaknesses	This project is specific for the Robert Bruce Thompson annual lighting design competition but can be adapted to other design contexts (e.g., chair, technology gadget).
Variants	Other consumer-ready AI and VR applications can be used (e.g., Adobe Firefly, Shape XR).
Assessment	Creative Product Semantic Scale (CPSS) ³ is used to assess student outcomes (10 AI sketches, 3 VR iterations, and one 3D printed prototype) in terms of creativity via three criteria: novelty, resolution, and style.

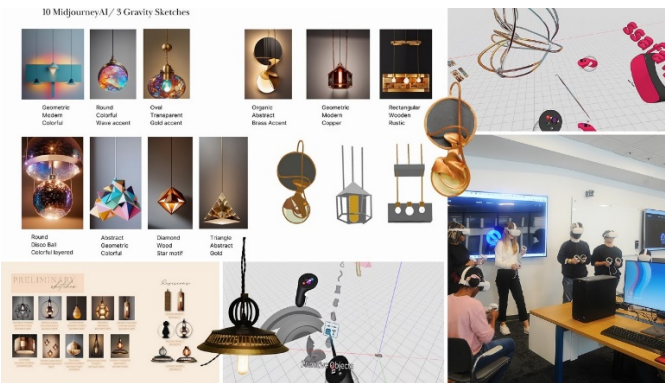


Figure 2: Sample student outcomes and classroom set up. Design by Sarah Sims and Maddie Fitch. Photograph by the author.

3 MATERIALS

The nine-week project includes:

- A Pro subscription of Midjourney AI (\$576 per year) that handles multiple prompts at a time (3 concurrent prompts, 10 queuing prompts). Students access the shared subscription via Discord from their laptops.
- A team subscription of Gravity Sketch VR (free for education with registration approvals). Stand-alone VR headsets such as Meta Quest 2.
- (Optional) Access to 3D printing such as a Fused Deposition Modeling (FDM) system using polylactic acid (PLA) filaments.

4 STUDENT OUTCOMES

The author assessed student outcomes with the abbreviated Creative Product Semantic Scale (CPSS) of 15 seven-point Likert semantic

pairs under three criteria: novelty (newness), resolution (function), and style (appearance). This is an established creativity measure with a strong internal consistency ($.72 \leq \alpha \leq .98$) [Vo, 2022, Wei et al., 2015]. The 2022 class showed significant statistical evidence for creativity improvement between AI ideation and VR iteration for all criteria: novelty, resolution, and style ($p < .05$). In a similar project conducted in 2021, the author only found moderate improvement in novelty and significant enhancement in style, between hand-sketching ideation and VR iteration [Vo, 2022]. AI and VR yielded a more comprehensive effect on student outcomes. Figure 2 presents exemplary student outcomes and the classroom setup.

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